CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Every formula and result of the chapter on one sheet

| | | |
|---|---|---|
| 1 | Cylinder: curved surface area | $2\pi rh$ |
| 2 | Cylinder: total surface area | $2\pi r(r+h)$ |
| 3 | Cylinder: volume | $\pi r^2 h$ |
| 4 | Hollow cylinder: volume of material | $\pi(R^2-r^2)h$ |
| 5 | Hollow cylinder: total surface area | $2\pi(R+r)h+2\pi(R^2-r^2)$ |
| 6 | Cone: slant height | $l=\sqrt{r^2+h^2}$ |
| 7 | Cone: curved surface area | $\pi rl$ |
| 8 | Cone: total surface area | $\pi r(l+r)$ |
| 9 | Cone: volume | $\frac{1}{3}\pi r^2 h$ |
| 10 | Sphere: surface area | $4\pi r^2$ |
| 11 | Sphere: volume | $\frac{4}{3}\pi r^3$ |
| 12 | Hemisphere | Curved $2\pi r^2$; total $3\pi r^2$; volume $\frac{2}{3}\pi r^3$ |
| 13 | Spherical shell: volume of material | $\frac{4}{3}\pi(R^3-r^3)$ |
| 14 | Recasting | Volume before $=$ volume after |
| 15 | Units | 1 litre $=1000$ cm³; 1 m³ $=1000$ litres |